

Simulation of Communication Systems using IT++

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In this article, we would like to introduce you to a versatile C++ library called IT++. The library is composed of a handful of useful classes and functions in mathematics, communications and signal processing. Academicians and industrialists around the world, working on communications, make extensive use of this library. The generic vector and matrix classes contained in the kernel resembles the approach used in MATLAB/Octave.

IT++ runs quite comfortably in multiple platforms including GNU/Linux, Sun Solaris, Microsoft Windows (with Cygwin, MinGW/MSYS or Microsoft Visual C++) and Mac OS X operating systems (OSes). The latest release of this software (version 4.3.1) is incorporated in this month's EFY Plus DVD.

Simulating communication systems

The main motivation behind the introduction of IT++ library is to provide users with the functionalities that help them in modelling communication systems.

Modulators. An analogue modulator transfers an analogue-baseband signal over an analogue-bandpass channel. Various analogue-modulation schemes like quadrature amplitude modulation (QAM) can be implemented using IT++.

The digital modulator transfers a digital bit stream over an analogue-bandpass channel. Binary phase shift keying (BPSK) and other digital modulation schemes are included in the software.

The pulse modulator transfers a narrowband-analogue signal over

```
#include <itpp/itcomm.h>
using namespace itpp;

//These lines are needed for use of cout and endl
using std::cout;
using std::endl;

int main()
{
    //Scalars
    int N;
    double N0;

    //Vectors
    bvec bits, dec_bits;
    vec symbols, rec;

    //Classes
    BPSK bpsk; //The BPSK modulator/demodulator class
    BERCC bercc; //The Bit Error Rate Counter class

    //Init
    N = 500000; //The number of bits to simulate
    N0 = 1; //0 dB SNR

    //Randomize the random number generator
    RNG_randomize();

    //Generate the bits:
    bits = randb(N);

    //Do the BPSK modulation
    bpsk.modulate_bits(bits, symbols);

    //Add the AWGN
    rec = symbols + sqrt(N0 / 2) * randn(N);
}
```

A sample code in IT++

Software at a Glance

Name of the tool: IT++

Type: C++ library for communication systems

Licence: GNU general-public licence

Latest release: Version 4.3.1

Website: itpp.sourceforge.net

Packages it can be interfaced with: Blas, Lapack and FFTW (optionally with Atlas, MKL and ACML)

a wideband-baseband channel. Schemes like pulse amplitude modulation (PAM) are provided in the library.

Vector modulators including orthogonal frequency division multiplexing (OFDM) and multiple input and multiple output (MIMO) are incorporated. The design of multi-user communication systems is aided by code division multiple access (CDMA) and OFDM modulation techniques.

Pulse shaping filters.

In order to adapt the transmitted waveform more to its original purpose or to make it suitable for the characteristics of the transmitting channel, waveform of the transmitting pulses needs to be altered. Raised cosine (RC) filter, a widely-used pulse-shaping filter with an inherent ability of minimising inter-symbol interference (ISI), could be implemented using IT++. Root raised cosine (RRC) filter is yet another pulse-shaping technique widely used in mobile communication.

Transmission through a channel.

Once a codeword representing the source message is formed, it can be transmitted through a channel. The channel could either be a binary symmetric channel (BSC) or additive white Gaussian noise (AWGN) channel. For transmission through a BSC, the effect of Gaussian noise is not considered. Instead, each bit received differs from the bit sent by an error probability, p .

When we consider AWGN channel, data received at each time equals data sent plus Gaussian noise with mean zero and some standard deviation.

Both these channels can be modelled using IT++.

Multi-path fading. When a signal is transmitted through a wireless medium, it propagates through multiple paths before reaching the antenna. Fading occurring in this path could either be frequency-selective or frequency-flat. If the channel's coher-